

Binomial expansion - Set 1 - Solutions

Specification MM2.4.

Answer key

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Worked solutions

Question 1 BIN-01 · Binomial expansion · spec MM2.4

Find the coefficient of x^3 in the expansion of $(2 + 3x)^5$.

Answer: B

$$\binom{5}{3}2^2(3x)^3 = 10 \times 4 \times 27x^3 = 1080x^3.$$

Full method - how and why

A single coefficient from $(a + bx)^n$. Use the general term rather than expanding the whole thing.

1 Write the general term: $\binom{5}{r}2^{5-r}(3x)^r$.

why: Every term of $(a + b)^n$ has the form $\binom{n}{r}a^{n-r}b^r$; the powers of the two parts always add to n .

2 For x^3 take $r = 3$: $\binom{5}{3}2^2(3x)^3$.

why: The power of x comes only from the second bracket, so r is the power you want.

3 Evaluate each factor: $\binom{5}{3} = 10$, $2^2 = 4$, $3^3 = 27$.

why: Keep them separate - combining too early is where slips happen.

4 Multiply: $10 \times 4 \times 27 = 1080$.

why: $10 \times 108 = 1080$.

Where it goes wrong

- Forgetting the 2^2 and answering 270.
- Using $3^3 = 9$ or cubing only the x , giving 360 - both are offered.

Takeaway The general term $\binom{n}{r}a^{n-r}b^r$ carries **three** factors: the binomial coefficient and a power of each part.

Question 2 BIN-02 · Binomial expansion · spec MM2.4

The coefficient of x^2 in the expansion of $(1 + kx)^6$ is 60. Given that $k > 0$, find k .

Answer: A

$$\binom{6}{2}k^2 = 15k^2 = 60, \text{ so } k^2 = 4 \text{ and } k = 2.$$

Full method - how and why

A coefficient is given and a constant inside the bracket is unknown. Set up the coefficient equation and solve.

1 General term of $(1 + kx)^6$: $\binom{6}{r}(kx)^r$, so the x^2 coefficient is $\binom{6}{2}k^2$.

why: The 1^{6-r} factor is invisible, which is why this form is the easiest to work with.

2 Compute $\binom{6}{2} = 15$, so the equation is $15k^2 = 60$.

why: $\binom{6}{2} = \frac{6 \times 5}{2} = 15$.

3 Solve: $k^2 = 4$, so $k = \pm 2$; the condition $k > 0$ gives $k = 2$.

why: Squaring means two roots; the stated condition selects one.

Where it goes wrong

- Forgetting to square k and solving $15k = 60$ to get $k = 4$ - which is offered.
- Using $\binom{6}{2} = 30$ (that is 6×5 without dividing by 2).

Takeaway The x^r coefficient of $(1 + kx)^n$ is $\binom{n}{r}k^r$ - the constant is raised to the same power as x .

Question 3 BIN-04 · Binomial expansion · spec MM2.4

What is the sum of the coefficients in the expansion of $(3x - 2)^7$?

Answer: D

Put $x = 1$: $(3 - 2)^7 = 1$.

Full method - how and why

Sum of all coefficients: substitute $x = 1$. No expansion needed at all.

- 1 Note that setting $x = 1$ makes every power of x equal to 1, so the expansion collapses to the sum of its coefficients.

why: If $(3x - 2)^7 = \sum c_k x^k$ then putting $x = 1$ gives $\sum c_k$.

- 2 Evaluate the original expression at $x = 1$: $(3 - 2)^7 = 1^7 = 1$.

why: You never need a single coefficient.

Where it goes wrong

- Expanding and trying to add eight terms under time pressure.
- Substituting $x = 0$, which gives only the constant term $(-2)^7 = -128$ - an offered option.

Takeaway $f(1)$ is the sum of the coefficients; $f(0)$ is the constant term; $f(-1)$ is the alternating sum.

Question 4 BIN-05 · Binomial expansion · spec MM2.4

Evaluate $\binom{10}{3} - \binom{10}{7} + \binom{9}{2}$.

Answer: A

$\binom{10}{3} = \binom{10}{7}$, so the first two cancel; $\binom{9}{2} = 36$.

Full method - how and why

A short calculation made trivial by the symmetry of binomial coefficients.

- 1 Use the symmetry $\binom{n}{r} = \binom{n}{n-r}$: since $3 + 7 = 10$, $\binom{10}{3} = \binom{10}{7}$.

why: Choosing 3 objects from 10 is the same as choosing which 7 to leave behind.

- 2 So the first two terms cancel, leaving $\binom{9}{2}$.

why: Spotting the cancellation avoids computing 120 twice.

- 3 Evaluate $\binom{9}{2} = \frac{9 \times 8}{2} = 36$.

why: $\binom{n}{2} = \frac{n(n-1)}{2}$.

Where it goes wrong

- Computing all three coefficients and mis-adding, e.g. $120 - 120 + 36$ done as 156.
- Thinking $\binom{10}{3} - \binom{10}{7}$ is $\binom{10}{-4}$ or similar.

Takeaway $\binom{n}{r} = \binom{n}{n-r}$: look for pairs whose lower entries add to n .

Question 5 BIN-06 · Binomial expansion · spec MM2.4

In the expansion of $(1+x)^n$, the coefficients of x^4 and x^5 are equal. Find n .

Answer: D

$\binom{n}{4} = \binom{n}{5}$ requires $4 + 5 = n$ (by symmetry $\binom{n}{r} = \binom{n}{n-r}$), so $n = 9$.

Full method - how and why

Two equal binomial coefficients from the same row. Symmetry gives the answer without computing factorials.

1 The condition is $\binom{n}{4} = \binom{n}{5}$.

why: In $(1+x)^n$ the coefficient of x^r is $\binom{n}{r}$.

2 Within one row of Pascal's triangle, two different entries are equal only when they are symmetric partners:

$\binom{n}{r} = \binom{n}{s}$ with $r \neq s$ forces $r + s = n$.

why: The row rises to a single peak and falls symmetrically, so equal values are mirror images.

3 Hence $n = 4 + 5 = 9$.

why: Check: $\binom{9}{4} = 126 = \binom{9}{5}$ ✓.

Where it goes wrong

- Setting up the factorial equation and mis-cancelling to get $n = 10$ or $n = 8$.
- Assuming $n = 5$ because the coefficients 'stop' there.

Takeaway Equal entries in the same row of Pascal's triangle are symmetric partners: $r + s = n$.

Question 6 BIN-07 · Binomial expansion · spec MM2.4

Expand $(1-2x)^4$ and hence find the coefficient of x^2 in $(1+3x)(1-2x)^4$.

Answer: D

$(1-2x)^4 = 1 - 8x + 24x^2 - 32x^3 + 16x^4$. Coefficient of x^2 in the product: $1 \cdot 24 + 3 \cdot (-8) = 0$.

Full method - how and why

A product of a binomial expansion with a linear factor. Only two products can give x^2 , so collect just those.

1 Expand $(1-2x)^4$: coefficients $\binom{4}{r}(-2)^r$ give $1 - 8x + 24x^2 - 32x^3 + 16x^4$.

why: $\binom{4}{1}(-2) = -8$ and $\binom{4}{2}(-2)^2 = 6 \times 4 = 24$; remember to raise the -2 to the power as well.

2 To make x^2 in the product, pair $1 \times 24x^2$ and $3x \times (-8x)$.

why: Degrees must add to 2: $(0, 2)$ and $(1, 1)$ are the only options.

3 Add: $24 + (-24) = 0$.

why: The two contributions cancel exactly - the answer is 0, not 'no term'.

Where it goes wrong

- Taking the x^2 coefficient of $(1-2x)^4$ as $+24$ but forgetting the $3x \times (-8x)$ contribution, giving 24.
- Sign error making $(-2)^2 = -4$.

Takeaway For a coefficient in a product, list the degree pairs that add to the target and sum their contributions.

Question 7 BIN-10 · Binomial expansion · spec MM2.4

Find the coefficient of x^5 in the expansion of $(1+x)^3(1+x)^4$.

Answer: B

$$(1+x)^3(1+x)^4 = (1+x)^7, \text{ and } \binom{7}{5} = 21.$$

Full method - how and why

Two binomials with the same bracket: combine the powers first, then use a single coefficient.

1 Combine: $(1+x)^3(1+x)^4 = (1+x)^7$.

why: Same base, so add the indices - this converts a product of expansions into one expansion.

2 The coefficient of x^5 is $\binom{7}{5}$.

why: Standard general term with $r = 5$.

3 Evaluate using symmetry: $\binom{7}{5} = \binom{7}{2} = \frac{7 \times 6}{2} = 21$.

why: Computing $\binom{7}{2}$ is quicker than $\binom{7}{5}$ and gives the same value.

Where it goes wrong

- Multiplying the separate expansions term by term and mis-collecting.
- Multiplying the coefficients $\binom{3}{5}$ and $\binom{4}{5}$ - both of which are zero, and neither is relevant.

Takeaway Combine equal brackets before expanding; then one binomial coefficient does the whole job.

Question 8 BIN-13 · Binomial expansion · spec MM2.4

Use the binomial expansion to find the value of 1.01^5 correct to 4 decimal places.

Answer: B

$$(1+0.01)^5 = 1 + 5(0.01) + 10(0.0001) + 10(0.000001) + \dots = 1 + 0.05 + 0.001 + 0.00001 + \dots \approx 1.0510.$$

Full method - how and why

An approximation with a small quantity. The binomial expansion converges fast because 0.01 is tiny.

1 Write $1.01^5 = (1+0.01)^5$ and expand: $1 + 5(0.01) + 10(0.01)^2 + 10(0.01)^3 + \dots$

why: Coefficients from row 5 of Pascal's triangle: 1, 5, 10, 10, 5, 1.

2 Evaluate the terms: $1 + 0.05 + 0.001 + 0.00001 + \dots$

why: $(0.01)^2 = 0.0001$, so the third term is $10 \times 0.0001 = 0.001$.

3 Add as far as the 4-decimal-place accuracy requires: $1.05101\dots \approx 1.0510$.

why: The fourth term only affects the fifth decimal place, so three terms suffice.

Where it goes wrong

- Stopping after two terms and answering 1.0500.
- Mis-scaling $(0.01)^2$ as 0.001, which gives 1.0600-style errors.

Takeaway With a small x , each binomial term is much smaller than the last - take just enough terms for the required accuracy.

Question 9 BIN-03 · Binomial expansion · spec MM2.4

Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

Answer: E

General term $\binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{12-3r}$. Need $12 - 3r = 0$, $r = 4$: $\binom{6}{4} \times 16 = 15 \times 16 = 240$.

Full method - how and why

'Term independent of x ' means the power of x is zero. Track the power of x in the general term and solve for r .

- General term: $\binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{12-2r-r}$.
why: $(x^2)^{6-r}$ gives x^{12-2r} and $\left(\frac{2}{x}\right)^r$ gives $2^r x^{-r}$; combine the powers.
- So the power of x is $12 - 3r$.
why: Getting this expression right is the whole question.
- Set it to zero: $12 - 3r = 0$, so $r = 4$.
why: 'Independent of x ' means x^0 .
- Evaluate the term: $\binom{6}{4} \times 2^4 = 15 \times 16 = 240$.
why: $\binom{6}{4} = \binom{6}{2} = 15$ by symmetry.

Where it goes wrong

- Using the power $12 - 2r$ and getting $r = 6$, giving $2^6 = 64$.
- Forgetting the 2^r factor and answering 15.

Takeaway Write the power of x in the general term as a linear expression in r , then set it to the value you need.

Question 10 BIN-08 · Binomial expansion · spec MM2.4

The first three terms in the expansion of $(1 + ax)^n$ in ascending powers of x are $1 + 12x + 54x^2$. Find n .

Answer: D

$na = 12$ and $\frac{n(n-1)}{2}a^2 = 54$. Dividing the second by the square of the first: $\frac{n-1}{2n} = \frac{54}{144} = \frac{3}{8}$, so $8n - 8 = 6n$ and $n = 4$ (then $a = 3$; check $\binom{4}{2} \cdot 9 = 54$).

Full method - how and why

Two coefficients, two unknowns (a and n). Dividing the equations cleverly eliminates a .

- Write the first two coefficients: $\binom{n}{1}a = na = 12$ and $\binom{n}{2}a^2 = \frac{n(n-1)}{2}a^2 = 54$.
why: The expansion of $(1 + ax)^n$ starts $1 + nax + \frac{n(n-1)}{2}a^2x^2$.
- Square the first equation: $n^2a^2 = 144$.
why: Squaring makes the power of a match the second equation, so dividing will cancel it.
- Divide the second by this: $\frac{n(n-1)a^2/2}{n^2a^2} = \frac{n-1}{2n} = \frac{54}{144} = \frac{3}{8}$.
why: Both a^2 and one factor of n cancel.
- Cross-multiply: $8(n-1) = 6n \Rightarrow 8n - 8 = 6n \Rightarrow n = 4$.
why: Then $a = 3$; check $\binom{4}{2} \times 9 = 6 \times 9 = 54$ ✓.

Where it goes wrong

- Dividing the second equation by the first (not its square), which leaves an a behind and stalls.
- Using $\binom{n}{2} = \frac{n^2}{2}$.

Takeaway To eliminate a constant that appears to different powers, raise one equation to the matching power first, then divide.

Question 11 BIN-09 · Binomial expansion · spec MM2.4

Which of the following statements is/are true?

- I** $\binom{n}{r} = \binom{n}{n-r}$ for all integers $0 \leq r \leq n$
II $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ for all integers $0 \leq r < n$
III $\sum_{r=0}^n \binom{n}{r} = 2^n$ for all positive integers n

Answer: H

I is the symmetry of Pascal's triangle; II is Pascal's rule (each entry is the sum of the two above it); III is $(1+1)^n$ expanded. All true.

Full method - how and why

Three standard properties of binomial coefficients. Each has a one-line reason and a one-line check.

- I:** $\binom{n}{r} = \binom{n}{n-r}$ - choosing r objects is the same as choosing the $n-r$ to leave. **True.**
why: Visible as the left-right symmetry of each row of Pascal's triangle.
- II:** $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ - Pascal's rule, i.e. each entry is the sum of the two above it. **True.**
why: Check with numbers: $\binom{3}{2} + \binom{3}{3} = 3 + 1 = 4 = \binom{4}{3}$ ✓.
- III:** $\sum_{r=0}^n \binom{n}{r} = 2^n$ - put $x = 1$ in $(1+x)^n$. **True.**
why: Equivalently, a set with n elements has 2^n subsets, counted by size.
- All three: 'I, II and III'.**
why: Match to the options.

Where it goes wrong

- Doubting Pascal's rule because of the index shift on the right - test it with small numbers.
- Thinking the sum in III is 2^{n+1} by miscounting the $n+1$ terms.

Takeaway Symmetry, Pascal's rule and row sum 2^n are the three facts about $\binom{n}{r}$ worth knowing cold.

Question 12 BIN-11 · Binomial expansion · spec MM2.4

Find the value of $(\sqrt{2} + 1)^4 + (\sqrt{2} - 1)^4$.

Answer: A

Odd powers of $\sqrt{2}$ cancel: $2[(\sqrt{2})^4 + \binom{4}{2}(\sqrt{2})^2 + 1] = 2[4 + 12 + 1] = 34$.

Full method - how and why

Adding conjugate expansions. The odd-power terms cancel, so you only need the even ones - and the surds disappear.

- Expand both with the binomial theorem. Their terms are identical except that odd powers of 1 and -1 alternate in sign.
why: $(\sqrt{2} + 1)^4$ and $(\sqrt{2} - 1)^4$ differ only in the sign of the odd-numbered terms.
- Adding doubles the even-power terms and cancels the odd ones: $2[(\sqrt{2})^4 + \binom{4}{2}(\sqrt{2})^2 + 1]$.
why: The terms kept are $r = 0, 2, 4$ in $\binom{4}{r}(\sqrt{2})^{4-r}$ - exactly those with an even power of $\sqrt{2}$, which are rational.
- Evaluate: $(\sqrt{2})^4 = 4$, $\binom{4}{2}(\sqrt{2})^2 = 6 \times 2 = 12$, and 1.
why: $(\sqrt{2})^2 = 2$ and $(\sqrt{2})^4 = 4$.
- Total: $2(4 + 12 + 1) = 2 \times 17 = 34$.
why: Rational, as expected - a sure sign the cancellation was done correctly.

Where it goes wrong

- Forgetting the factor 2 and answering 17.
- Keeping a surd term and choosing $34 + 24\sqrt{2}$, which is $(\sqrt{2} + 1)^4$ alone.

Takeaway Adding conjugate binomials kills the odd terms, leaving a rational answer - expand only the even ones.